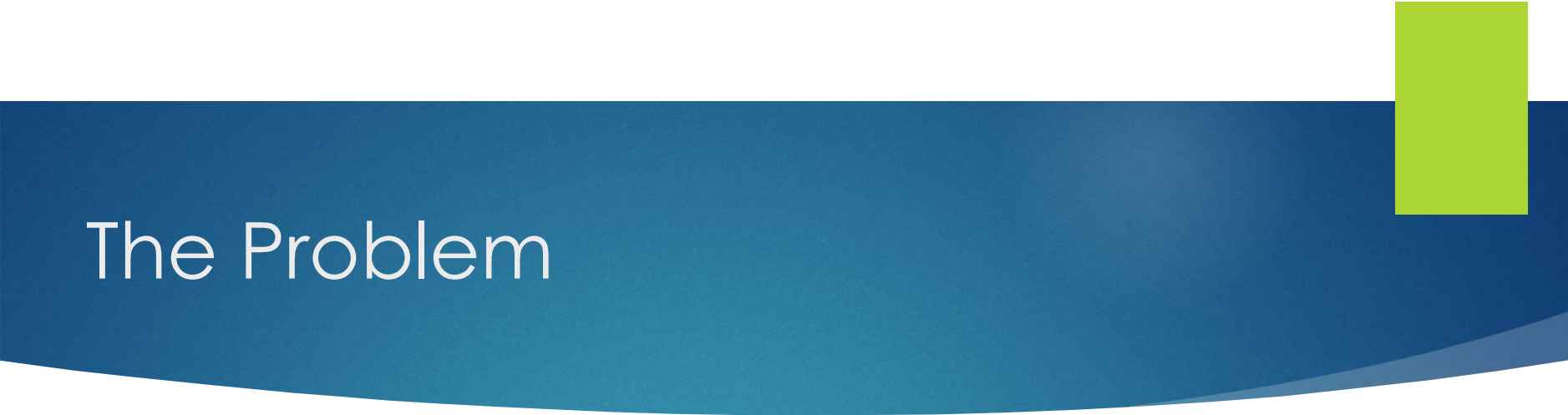




SES-2: Strategic Energy Shield System

Founder: Spencer O'Brien

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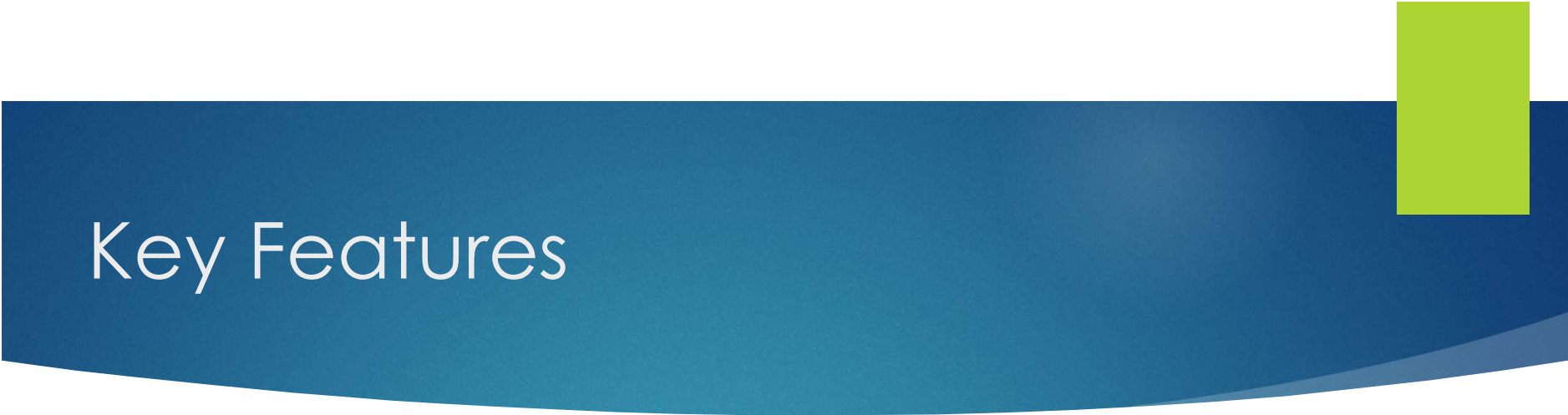
The Problem

- Growing global energy demand
- Grid vulnerability to geopolitical, climate, and cyber risks
- Limited scalability of terrestrial renewables
- Urgent need for resilient, clean, uninterrupted power



The Solution – SES-2

- SES-2 is a rectangular, modular, AI-managed orbital power system with beamforming and in-orbit construction
- Collects solar energy in geostationary orbit
- Beams energy wirelessly to Earth-based receivers via beamforming
- Resilient, scalable, autonomous, and secure
- 20 frontier technologies powering SES, including modular robotics, AI predictive routing, phased-array transmission, blockchain-verified energy tracking
- Beam tokenization platform for energy credit trading



Key Features

- Origami-folding ultrathin photovoltaic panels (CIGS, perovskite, etc.)
- Semi-autonomous deployment & maintenance via robotic swarm
- AI-driven adaptive beamforming to Earth or Mars receivers
- Orbital mesh layer with quantum encryption and debris-clearing lasers
- Modular growth: from pilot array to global infrastructure

How do we get supplies to the SES for expansion?

Strategy

Ultralight folded solar film (<1 kg/m²)

Origami folded supermodules

Orbital assembly bots

3D printed trusses

On-orbit unspooling systems

AI auto-alignment & tensioning

Effect

20-30x compaction

1 launch = 4-5 km²

Expand small payloads into large arrays

Replaces heavy mechanical scaffolding

Launch compact rolls, deploy in orbit

Speeds up autonomous buildout

This will allow us to achieve 1000 Starships launches for entire SES-2 system.



Deployment Phases

- Phase 1: Pilot launch (1–5 km²)
- Phase 2: Scaled growth with robotic in-orbit assembly
- Phase 3: Global-scale array (targeting 4178 km², scaling toward Rhode Island-sized platform)
- On-orbit manufacturing hubs for maintenance and expansion



Use Cases & Impact

- Power for critical infrastructure, AI, and defense systems
- Emergency backup in disaster zones
- Secure energy for allied governments & space missions
- Free power for bottom 5% GDP nations; dual-use as solar radiation filter
- Beam rerouting protocols prevent airspace conflicts



Competitive Advantage

- First truly autonomous, scalable space-based energy platform
- U.S.-controlled with allied partner assist
- Patent-pending rectangular modular design, AI-routing, and in-orbit construction system
- No dependence on terrestrial limitations
- Can route power globally with precision and resilience



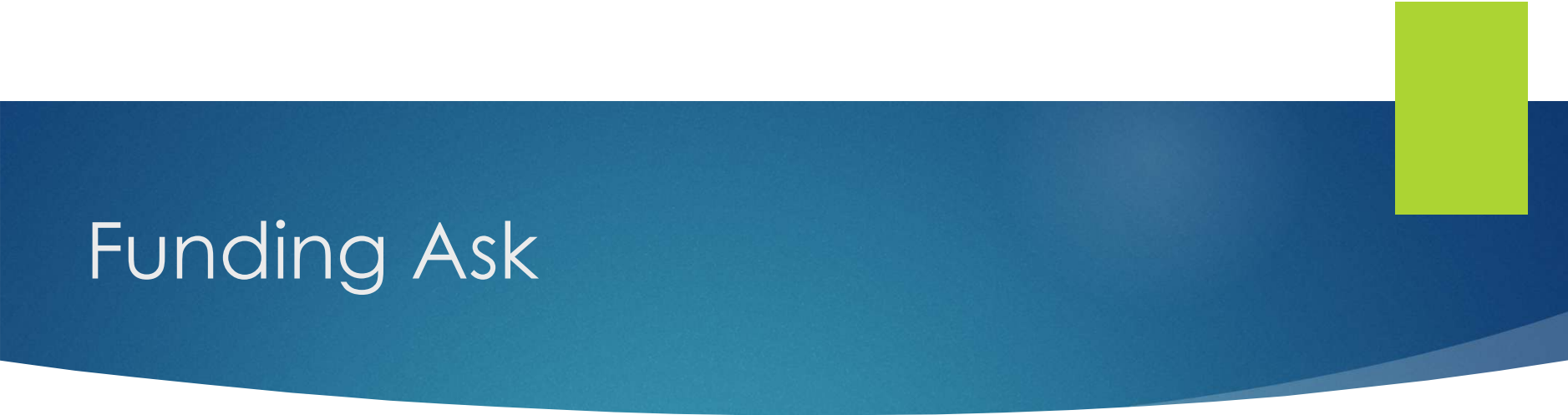
Business Model

- Tiered energy pricing by usage class and urgency
- Long-term contracts with governments and hyperscalers
- Premium rate for mobile or on-demand energy delivery
- Licensing of manufacturing hubs, swarm logic, and AI-based energy tokenization



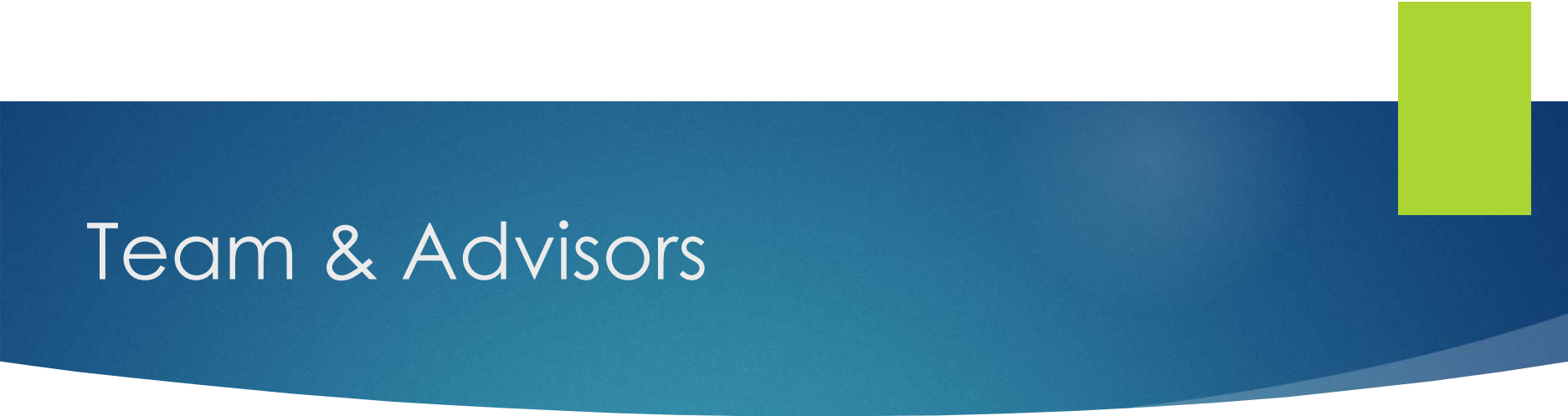
Market Opportunity

- Global energy market: \$10 trillion/year
- Growing demand for secure, off-grid, and sustainable energy
- Critical infrastructure & AI power use accelerating



Funding Ask

- Seeking \$2.5M seed funding (plus follow-on to \$10M Series A)
- Use of funds: orbital simulation, prototype panel development, space law/IP support, and partner engagement
- Roadmap to Series A includes test array and ground receiver trials



Team & Advisors

- Founder and vision lead: Spencer O'Brien
- Strategic advisors (space, energy, defense, AI)
- Partners: [TBD – open to SpaceX, NASA, DoD, etc.]



Vision & Closing

- SES is the future of resilient, space-based global energy
- Join us in building the orbital energy grid of tomorrow
- Contact: spencer@nocturnalsecurity.io | www.nocturnalsecurity.io